DO NOW — RECALL

Karnaugh maps 2: simplify to gates

Name: _____ Class: _____ Date: _____

1. In a Karnaugh-map pair, A stays 0 and B changes between 0 and 1. What is the simplified term?

2. Show the array after one complete left-to-right pass of ascending bubble sort on [8, 3, 6, 2] .

3. C#: int[,,] scores = new int[2, 3, 4]; How many values can it store? What does scores.GetLength(2) return?

4. Add 01000000 0011 and 01000000 0010 . Format: 8-bit two’s-complement fractional mantissa + 4-bit two’s-complement exponent . Show exponent alignment and the final normalised result.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

¬A A stays 0, so retain NOT A (¬A). B changes, so remove B.

2. Model answer

[3, 6, 2, 8] Swap 8 and 3 → [3, 8, 6, 2]; swap 8 and 6 → [3, 6, 8, 2]; swap 8 and 2 → [3, 6, 2, 8]. The largest value is now last.

3. Model answer

24 values; GetLength(2) = 4 2 layers × 3 rows × 4 columns = 24 values. Dimension 2 is the third dimension (columns).

4. Model answer

01100000 0011 (denary 6) Align to exponent 3 by shifting the second mantissa right: 01000000 → 00100000. Add the mantissas: 01000000 + 00100000 = 01100000. The exponent remains 0011. This is normalised: $0.75 \times 2^3 = 6$.

By the end, you can…

- choose efficient K-map groups including wraparound and overlap
- derive a simplified expression from each group
- translate the simplified expression into an equivalent gate plan

Vocabulary — know these terms

Recall the term — efficient group

Recall the term — overlap

Recall the term — wraparound

Recall the term — literal

Recall the term — product term

Recall the term — simplified expression

Recall the term — equivalent

Recall the term — gate plan

NEW LEARNING

Read, annotate and explain

Start with the map

Find the simplest rule for these outputs.

Method: Circle → Compare → Remove changing inputs → Write expression → Draw gates.

Find the 1s

Look at the 2×2 map on the left. Three squares contain 1 . We need to cover them in groups of neighbouring 1s.

Use rectangles containing 1, 2, 4, 8 or 16 cells. Include only 1s, and choose the largest useful groups. Diagonal cells are not neighbours.

A square may belong to two groups

This is overlap . It is allowed when it makes the expression simpler.

Keep the inputs that stay the same

Bottom pair: A stays 1; B changes. Keep A.

Right pair: B stays 1; A changes. Keep B.

The outside columns form one group

BC runs 00, 01, 11, 10 . The first and last columns are neighbours: only B changes between 00 and 10 .

Imagine rolling the map so its left and right edges meet. The four 1s make a 2 × 2 group ; the middle 0s are outside it.

Compare all four cells

A changes between the two rows → remove A.

B changes between columns 00 and 10 → remove B.

C stays 0 in all four cells → keep NOT C.

Gate plan: C → NOT gate → Q.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

How do two groups become a circuit?

Find the two pairs of 1s

ORANGE · TOP: circle columns 01 and 11 where A = 0.

BLUE · BOTTOM: circle columns 11 and 10 where A = 1.

Follow the same coloured groups on the map on the left. Each pair becomes one term.

Keep ¬A and C

A = 0 in both squares, so write ¬A . C = 1 in both, so keep C . B changes , so remove B.

Keep A and B

A = 1 in both squares, so keep A . B = 1 in both, so keep B . C changes , so remove C.

Join the two pairs with OR

Either coloured pair makes Q = 1 . AND the inputs inside each pair; OR the two results together.

Draw the actual logic gates

Trace it: A goes through NOT for the orange pair; A also goes directly to the blue pair. Each pair feeds an AND gate. OR the two AND outputs to get Q.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Find one pair

Method / working:

2. Let two groups share a 1

Method / working:

3. Remember the outside edges touch

Method / working:

4. Try a bigger Karnaugh map

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

1. Find one pair
Circle the largest valid groups of 1s.
Compare the inputs that remain constant in each group.
Simplify to the Boolean expression Q.
Draw the logic gates and connections.

Question 2

2. Let two groups share a 1
Circle the largest valid groups of 1s.
Compare the inputs that remain constant in each group.
Simplify to the Boolean expression Q.
Draw the logic gates and connections.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

3. Wrap around the outside edges
Circle the largest valid groups of 1s.
Compare the inputs that remain constant in each group.
Simplify to the Boolean expression Q.
Draw the logic gates and connections.

Question 4

4. Two groups in a bigger map
Circle the largest valid groups of 1s.
Compare the inputs that remain constant in each group.
Simplify to the Boolean expression P.
Draw the logic gates and connections.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A K-map simplification is valid only if the simplified expression gives the same output for every possible input combination.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.